

Airbnb Data Science Analysis Report

Comprehensive Machine Learning and Deep Learning Study

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EXECUTIVE SUMMARY

This comprehensive analysis of Airbnb listing data demonstrates a complete data science pipeline from exploratory data analysis through advanced machine learning and deep learning implementations. The project successfully developed production-ready models for price prediction and popularity classification, with XGBoost achieving 9.7% prediction error for pricing recommendations.

1. EXPLORATORY DATA ANALYSIS (EDA)

Dataset Overview

The dataset used contained a total of 419 Airbnb listings with 16 original columns including price, location, property details, and host information. Target variables were Price (for regression) and Popularity classification (based on reviews). Listings spanned multiple cities, each identified by latitude and longitude coordinates.

Key Data Quality Findings

Missing values were minimal (<5%) across most features. The dataset contained a mix of numerical and categorical data, with prices ranging from \$24 to \$1,050 and a median around \$122. Property types included entire homes, private rooms, and shared rooms.

Critical EDA Insights

The price distribution was right-skewed, with most listings priced between \$50–\$200. Outliers above \$500 represented luxury segments; log transformation was applied to stabilize model performance.

Geographically, listings clustered near city centers, and higher-priced areas correlated strongly with latitude and longitude. Accommodation capacity proved to be the strongest predictor of price, with each additional guest increasing the price by roughly \$25. Host experience also showed a positive relationship with popularity and pricing, particularly for professional hosts managing multiple listings. Feature engineering was guided by these observations, leading to creation of new variables such as amenities_count and host_experience_years.

Data Preprocessing Decisions

Outlier treatment was carefully performed by retaining luxury listings for market representation. Label encoding was applied for categorical features, and scaling (StandardScaler) was used for neural network models. An 80/20 train-test split was implemented to maintain balanced distributions.

2. FEATURE ENGINEERING

Feature engineering expanded the dataset from 16 to 22 clean, non-leaky features. Notable new features included `host_experience_years`, `amenities_count`, `has_reviews`, and encoded versions of `property_type`, `room_type`, and `stay_type`. To prevent data leakage, review-based and time-dependent variables were removed (e.g., `number_of_reviews`, `review_scores_rating`, `last_review`). Scaling was tailored per model type: `StandardScaler` for linear and neural models, while tree models used unscaled data.

3. REGRESSION MODELING (PRICE PREDICTION)

Four models were developed: Linear Regression, Random Forest, XGBoost, and Neural Networks. Linear Regression established a baseline with \$27.84 MAE (22.7% error), while Random Forest reduced the error to 15.8%. Tuned XGBoost achieved the best result—\$11.85 MAE (9.7% error) and $R^2=0.885$ —making it production-ready. Feature importance highlighted 'accommodates' (34.2%) and 'latitude' (15.8%) as dominant predictors. Business implications include more accurate pricing strategies, optimized host revenue, and competitive marketplace positioning.

4. CLASSIFICATION MODELING (POPULARITY PREDICTION)

A binary classification target defined listings as 'popular' or 'not popular' based on median review counts. After removing leakage features, Random Forest achieved 83.3% accuracy and 0.906 AUC, outperforming Logistic Regression. Key predictors included `host_experience_years`, `minimum_nights`, and `amenities_count`. From a business perspective, minimizing false negatives is critical as it prevents missed revenue opportunities from potentially popular listings.

5. DEEP LEARNING EXPERIMENT

TensorFlow installation faced technical challenges due to environment conflicts, so sklearn's `MLPRegressor` was used as a neural network alternative. The tuned network achieved significant improvement—reducing MAE from \$37.69 to \$15.84 (12.9% error). Despite limited data (335 samples), it delivered competitive performance second only to XGBoost. The study highlights that deep learning can rival classical models when optimized, even on moderate datasets.

6. FINAL MODEL COMPARISON

XGBoost emerged as the top model (9.7% error), followed by the tuned neural network (12.9% error) and Random Forest (15.8% error).

Linear Regression remained a transparent baseline. The production recommendation is XGBoost for price prediction due to its balance of accuracy and efficiency. For classification tasks, Random Forest offers the best balance between interpretability and performance.

7. BUSINESS RECOMMENDATIONS

Deploy XGBoost for automated pricing recommendations and Random Forest for targeted marketing campaigns. Emphasize property size and host experience as major determinants of price and popularity. Encourage hosts to enhance amenities and adopt flexible booking policies. The study validated that size, location, and experience are core factors driving Airbnb success. Technical achievements include identifying and resolving data leakage, executing robust cross-validation, and conducting comprehensive hyperparameter tuning.

8. GIT WORKFLOW DOCUMENTATION

This section is reserved for GitHub workflow visuals, demonstrating repository creation, branch management, commit history, and pull requests.

The screenshot shows a GitHub repository page for 'Buildable-ML-DL-Fellowship / Week_7'. The repository is owned by 'juni2003' and has a description 'Complete data preparation and feature engineering for Week 7'. The current branch is 'main', and the repository is 3 commits ahead of the upstream 'main'. The file list shows two files: 'Airbnb_Complete_Analysis_EDA_FE.ipynb' and 'README.md'. The 'README.md' file is selected, showing its content: 'Week 7: Airbnb Listing Analysis (Albany, NY)' and 'This folder contains all code, notebooks, figures, and reports for the Week 7 assignment of the Buildable ML/DL Fellowship Program.' Below the README content is a section titled 'Assignment Overview'.

Name	Last commit message	Last commit date
..		
Airbnb_Complete_Analysis_EDA_FE.ipynb	Complete data preparation and feature engineering for Week 7	now
README.md	Add README for Week 7 Airbnb Listing Analysis	yesterday

Week 7: Airbnb Listing Analysis (Albany, NY)

This folder contains all code, notebooks, figures, and reports for the Week 7 assignment of the Buildable ML/DL Fellowship Program.

Assignment Overview

Complete data preparation and feature engineering for Week 7 #4

Merged

juni2003 merged 1 commit into week7-regression-classification from week7-assignment now

Conversation 0

Commits 1

Checks 0

Files changed 1

+1,869 -0

juni2003

commented 1 minute ago

Owner

...

Bring EDA and Feature Engineering work to regression branch

- Dataset: 419 Airbnb listings with comprehensive EDA
- Features: 26 engineered features across 5 categories
- Train/test split: 335/84 with stratification ready
- Targets: price regression + popularity classification
- Ready for modeling phase

📄

Complete data preparation and feature engineering for Week 7

Verified

bbd3883

juni2003 merged commit 893db46 into week7-regression-classification now

Revert

Pull request successfully merged and closed

You're all set — the week7-assignment branch can be safely deleted.

Delete branch

Successfully merging this pull request may close these

Reviewers

Suggestions

Copilot

Request

Still in progress?

Convert to draft

Assignees

No one — assign yourself

Labels

None yet

Projects

None yet

Milestone

No milestone

Development

juni2003 / Buildable-ML-DL-Fellowship

Type / to search

Code

Issues

Pull requests

Actions

Projects

Wiki

Security

Insights

Settings

Files

week7-regression-cl... + 🔍

Go to file

Week_1

Week_3

Week_4

Week_6

Week_7

Airbnb_Complete_Analysis_EDA...

Airbnb_Regression_Classification...

README.md

README.md

Buildable-ML-DL-Fellowship / Week_7 /

Add file + 📄 ...

juni2003 Complete regression and classification modeling with full results c4be25a · 3 minutes ago History

This branch is 5 commits ahead of main .

Contribute

Name	Last commit message	Last commit date
..		
Airbnb_Complete_Analysis_EDA_FE.ipynb	Complete data preparation and feature engineering for Week 7	2 hours ago
Airbnb_Regression_Classification_Modeling.ipynb	Complete regression and classification modeling with full results	3 minutes ago
README.md	Add README for Week 7 Airbnb Listing Analysis	yesterday

README.md

✎

☰

Week 7: Airbnb Listing Analysis (Albany, NY)

This folder contains all code, notebooks, figures, and reports for the Week 7 assignment of the Buildable ML/DL Fellowship Program.

Open

Integrate ML models: Regression and Classification implementations #5

juni2003 wants to merge 2 commits into `week7-assignment` from `week7-regression-c...`

BUSINESS IMPACT:

- XGBoost enables accurate pricing strategies (9.7% error)
- Random Forest supports targeted marketing (83% accuracy)
- Feature insights guide host improvement recommendations

Ready for: Deep learning experiment integration
Branch status: Feature-complete classical ML pipeline

👤

juni2003 added 2 commits 1 hour ago

🔗

Merge pull request #4 from juni2003/week7-assignment

Verified893db46

🔗

Complete regression and classification modeling with full results

Verifiedc4be25a

🔗

No conflicts with base branch

Merging can be performed automatically.

Merge pull request

You can also merge this with the command line. [View command line instructions.](#)

Notifications

Customize

Unsubscribe

You're receiving notifications because you authored the thread.

1 participant

🔒 Lock conversation

Merged

Integrate ML models: Regression and Classification implementations #5

juni2003 merged 2 commits into `week7-assignment` from `week7-regression-c...` now

5-fold cross-validation implemented

Comprehensive model comparison with business metrics

Professional visualizations for model interpretation

BUSINESS IMPACT:

- XGBoost enables accurate pricing strategies (9.7% error)
- Random Forest supports targeted marketing (83% accuracy)
- Feature insights guide host improvement recommendations

Ready for: Deep learning experiment integration
Branch status: Feature-complete classical ML pipeline

👤

juni2003 added 2 commits 1 hour ago

🔗

Merge pull request #4 from juni2003/week7-assignment

Verified893db46

🔗

Complete regression and classification modeling with full results

Verifiedc4be25a

🔗

juni2003 merged commit e9429d3 into `week7-assignment` now

Revert

🔗

Pull request successfully merged and closed

You're all set — the `week7-regression-classification` branch can be safely deleted.

Delete branch

Successfully merging this pull request may close these issues.

None yet

Notifications

Customize

Unsubscribe

You're receiving notifications because you modified the open/close state.

1 participant

🔒 Lock conversation

9. PROJECT COMPLETION STATUS

- EDA: 15+ visualizations completed with comprehensive insights.
- Feature Engineering: 22 clean features, no leakage.
- Regression: XGBoost achieves production-ready 9.7% error.
- Classification: Random Forest achieves 83.3% accuracy.
- Deep Learning: Tuned neural network competitive and efficient.
- Documentation: Full analysis with actionable business recommendations.